

Question				Marking details	Marks available				Maths	Prac
					AO1	AO2	AO3	Total		
	(b)	(i)		$R = \frac{V^2}{P} \text{ or } \left[I = \frac{1000 \text{ W}}{230 \text{ V}} \text{ and } R = \frac{V}{I} \right] (1)$ $R = 53 \Omega \text{ [accept early rounding]} (1)$	1	1		2	2	
		(ii)		3.6 [MJ]		1		1	1	
	(c)			If electric heating is used rather than gas, for a given heating effect more <u>CO₂</u> produced [or more gas, a non-renewable resource [accept: fossil fuel] is used]. (1) Greater contribution to climate change/global warming [so use of electric heaters <i>should</i> be discouraged] (1) Any one of... • [But] higher cost of electricity [per kW] is discouragement enough • [But] not all power stations <i>are</i> gas-fired. [Some eg nuclear, wind, don't produce CO₂] • [But] not all homes have gas available • CO₂ absorbs long-wavelength infrared (or infrared emitted from Earth's surface) in context of climate change • Any other relevant and non-trivial point e.g. safety issue with naked flames / <u>CO</u> emission / gas leaks Example of point not worth credit: "it's none of anyone else's business how I heat my home" – the word in the question was "discouraged" / electric heaters more expensive to run			3	3		
				Question 1 total	5	7	3	15	7	0